

System Design Document

furlink

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Introduction

This System Design Document outlines the technical architecture for furlink. furlink is designed to replace manual booking process with an integrated digital ecosystem. The system centralizes pet registration, scheduling, and payment processing into a single platform accessible via web.

Purpose

The purpose of this document is to ensure furlink design meets established project requirements for real-time booking and business management. It provides a detailed description of the system architecture, software modules, and security protocols to ensure organizational goals for efficient pet care management are achieved.

System Overview

This section describes the core goals and approach of furlink. Unlike manual methods that led to scheduling conflicts and inconsistent management, furlink standardizes the booking experience to improve efficiency for both pet owners and service providers.

The system is designed to achieve four primary goals:

1. **Reduce Scheduling Conflicts:** Minimize double booking by ensuring that each service provider does not experience booking incidents during the first three (3) months of the release.
2. **Improve Visibility of Pet Service Providers:** Improve the visibility of small to medium-sized pet service providers by giving them a dedicated platform to showcase their services and directly connect with pet owners. Onboard and verify one (1) service provider within the first three (3) months of release, and each provider receives search engine optimization (SEO) profile dashboard to improve visibility and potential customer reach.

3. **Streamline the Booking Journey:** Achieve 95% accuracy in understanding user requests and automate 50% of data entry for registrations and bookings. Within the first three (3) months, this will eliminate manual errors and speed up the delivery of PayMongo payment links.
4. **Provide Business Insights for Service Providers:** Enable pet service providers to track monthly sales, customer feedback, and related insights by integrating a business analytics dashboard into the platform by August 3, 2026.

furlink utilizes a single shared database to ensure information remains synchronized and up to date. The website is developed using React.js, with Supabase (PostgreSQL) serving as the backend and database. PayMongo is integrated as the secure payment gateway, and Supabase Edge Functions handle complex background tasks such as AI processing, notifications, and payment webhooks.

Design Constraints

The team identified several constraints which will impact and limit the design of furlink. These constraints are beyond the scope of furlink but must be carefully factored into the system design. To date, the following constraints have been identified:

1. Hardware and Technology Constraints

- **Constant Internet Requirement:** The system is purely cloud-based and requires a steady internet connection for all core functions including listing, booking, payments, and data synchronization.
- **Serverless Cold Start Delays:** Serverless Edge Functions may cause delays of 1 to 3 seconds during periods of inactivity, affecting AI processing and notification delivery.
- **AI Rate Limiting:** The Gemini 2.5 Flash will still have limited request per minute and request per day. During peak periods, concurrent users interacting with the AI Booking Assistant may encounter delayed responses if their limits are reached.

2. Software and System Integration

- **Payment Webhook Dependency:** Booking finalization is entirely dependent on receiving a payment confirmation signal from PayMongo. Any delay or failure in webhook delivery directly blocks the booking workflow.
- **Role-Based Access Complexity:** The system supports three distinct user roles – pet owners, service providers, and application administrator – each requiring separate permission sets that must be carefully maintained across all modules.
- **AI Context Window Dependency:** The AI Booking Assistant relies on Gemini's context window to retain pet and booking details throughout a conversation. Conversation that exceeds the session context may require the user to re-provide information.

3. Standards and Legal Rules

- **Data Privacy Compliance:** The system must comply with the Data Privacy Act of 2012 or Republic Act 10173 for all stored personal, medical, and business records, including the right to erasure upon formal request. The AI Booking Assistant is constrained to only access pet and booking data that the user has explicitly provided or already stored in their profile to remain compliant with the Data Privacy Act of 2012.
- **Payment Handover:** The AI Booking Assistant cannot collect or store card details at any point. It is strictly limited to generating PayMongo checkout link, with the final transaction always completed on PayMongo's secure external page.
- **Pet Care Operational Workflows:** The platforms must support industry-standard pet care processes such as vaccination record submission and illness history verification.
- **Audit Accountability:** An audit trail must be maintained across all transactions to ensure transparency and accountability in line with service industry standards.

Roles and Responsibilities

The table below defines furlink System Design roles and responsibilities and serves as the list of points of contact for issues and concerns:

Name	Role	Email
Lalainne Andaya	Project Manager & Developer	lcandaya2@student.apc.edu.ph
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Project References

furlink is designed in accordance with several market studies, industry analyses, consumer surveys, and findings from existing booking platforms. The following references inform the system's design decisions should be referred to individually for more detailed information.

Pet Care Industry

- Philippine Pet Care Market Report, 2025
- Vero ASEAN Consumer Behavior Survey, 2024
- Southeast Asian Pet Care Market Projection Report
- Metro Manila Pet Service Provider Survey
- Post-Pandemic Philippine Pet Care Growth Study (Klarence et al.)

Online Booking Systems

- Checkfront
- SimplyBook.me

- Zoho Bookings
- ADB Digital Commerce Report, 2022
- Grab, Agoda, and Foodpanda Platform Case Studies

Technology in Pet Care Bookings

- Salesforce
- Infobip
- Bloomreach
- PetHealthHub Platform Review
- PetBacker Platform Analysis
- Rover Platform Review

System Architecture

The architecture necessary to achieve the system design for furlink consists of following hardware and software components:

Hardware

furlink is a fully cloud-based system and does not rely on any physical or on-premises hardware. All infrastructure is managed through the following third-party cloud providers:

- Vercel: Hosts and deploys the web application with automated builds and continuous deployment connected to the GitHub Repository.
- Supabase: Manages all database instances, authentication services, and file storage on cloud infrastructure.
- GitHub Codespaces: Provides the team's cloud-based development environment.
- User Devices: Accessible from any modern browser-capable device. No proprietary hardware is required.

Software

Frontend

- React.js: Primary framework for user interface rendering and client-side logic.
- Vite: Build tool providing fast development and optimized production builds.
- Vercel AI SDK: Client-side library that enables the React frontend to communicate with the Gemini AI model and handle tool-calling responses for form autofill and data retrieval.

Backend

- Supabase Auth with JSON Web Token (JWT): Manages authentication and session security.
- Supabase Row Level Security (RLS): Enforces data access boundaries at the database level.
- Supabase Edge Functions: Handles serverless background logic including webhooks, AI orchestration, and automated triggers. The Gemini API key is stored securely as a Supabase secret and is never exposed to the client.
- Gmail SMTP: Delivers transactional email notifications triggered by Edge Functions.

Payments

- PayMongo: Payment gateway integrated via API for booking payment processing and webhook-based confirmation.

Artificial Intelligence

- Google Gemini 2.5 Flash: The AI model powering the AI Booking Assistant. Selected for its high-speed conversational performance, large context window suitable for retaining pet and booking details across long conversations.
- Vercel AI SDK with @ai-sdk/google provider: Orchestrates communication between the React frontend and the Gemini model, enabling tool calling for database queries and structured data extraction for form autofill.

Version Control and Collaboration

- GitHub Repository: Central codebase with version control and Vercel integration for continuous deployment.
- GitHub Codespaces: Cloud-hosted development environment for the team.

Software Modules

The following modules operate through the shared Supabase backend:

- Authentication and Access Control Module
- User Profile Management Module
- Pet Profile Module
- Service Provider Onboarding Module
- Platform Admin Oversight and Moderation Module
- Transaction and Payment Module
- Provider Operations Module
- Cancellation and Rescheduling Module
- Business Intelligence Dashboard Module
- AI Booking Assistant Module
- Discovery and Filtering Module

Database Design

furlink's database is hosted on Supabase, a cloud-based platform built in PostgreSQL. It serves as the single centralized repository for all system data, ensuring that information entered through any part of the platform is immediately consistent and accessible across all modules.

All tables use UUID primary keys and are governed by Row Level Security (RLS), which enforces data access rules directly at the database level. Authenticated users may only read, insert, update, or delete records within the boundaries of their assigned role. This applies independently of the frontend, and unauthorized operations are rejected at the database level even if a request bypasses the application.

The database is organized into seven logical groups corresponding to the platform's core functional areas:

- Identity and Access
- Pet Management
- Service Provider
- Services and Pricing
- Booking Flow
- Post-Service
- Notifications

Each group contains tables that are directly linked through foreign key relationships to maintain referential integrity across the system.

Several automated behaviors are enforced at the database level through PostgreSQL triggers and Edge Functions, including real-time email and in-app notifications, automatic booking decline after 24 hours of no service provider response, and automatic booking completion 3 days after the scheduled date to unlock the review window. The AI Booking Assistant also interacts with the database through secured Edge Functions, querying the tables for availability and information searches and writing when auto-filling and confirming bookings.

Hardware and Software Detailed Design

Hardware

furlink does not utilize any physical or on-premises hardware since all infrastructure is managed through cloud service providers as documented in the System Architecture section.

Software

furlink's software is organized into modules that each handle a distinct area of platform functionality. The design was developed around the specific needs of pet owners, service providers, and the platform administrator, and was compartmentalized into modules to

allow independent development, testing, and maintenance. All modules communicate through the shared Supabase backend and are supported by separately prepared system diagrams for more detailed technical reference.

- **Authentication and Access Control Module:** Manages registration, login, and role assignment for three user types through Supabase Authentication and JWT, with permissions enforced through RBAC and RLS.
- **User Profile Management Module:** Allows users to update personal details, change passwords, view their role, and deactivate their account via soft-delete mechanism.
- **Pet Profile Module:** Enables pet owners to create and maintain permanent pet profiles that can be referenced during the booking process to reduce data entry.
- **Service Provider Onboarding Module:** An application workflow collecting business credentials, service offerings, pricing, staff, operating hours, and supporting documents for application admin review.
- **Platform Admin Oversight and Moderation Module:** Provides the application administrator with tools to approve or reject service provider applications and manage user warnings or suspensions.
- **Transaction and Payment Module:** Initiates a PayMongo checkout session upon booking, redirects the user to PayMongo's hosted page, and finalizes the booking only upon receiving a verified webhook confirmation. furlink does not handle raw card details at any point.
- **Provider Operations Module:** Gives service providers a real-time interface to accept or decline booking requests, with status changes reflected instantly on the pet owner's appointment page and booking history.
- **Cancellation and Rescheduling Module:** Handles cancellations, rescheduling requests, and post-service review submission, with review eligibility automatically unlocked three days after the scheduled booking date.
- **Business Intelligence Dashboard Module:** Presents service providers with visual summaries of business performance, sales performance, and customer insights.

- **AI Booking Assistant Module:** A conversational assistant powered by Google Gemini 2.5 Flash that automates and simplifies the booking process through three core capabilities:
 - Knowledge Retrieval that allows the assistant to search system data including service listings, provider availability, and general FAQs by triggering secured Edge Functions that query the Supabase database, then summarizing result in plain language for the user.
 - Automated Data Entry that allows the assistant to extract structured pet and booking details from natural conversation and automatically populate the Pet Registration and Booking Forms, allowing the user to simply review and confirm.
 - Payment Orchestration that allows the assistant to trigger a final Edge Function that call the PayMongo API to generate a unique checkout link once the user confirms booking details, which is then presented in the chat for the user to complete payment. The booking is marked as confirmed in Supabase only after a PayMongo webhook verifies successful payment.
- **Discovery and Filtering Module:** Allows pet owners to search and filter service providers by location, star rating, and price range.

System Security and Integrity Controls

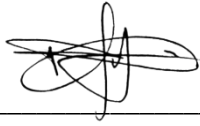
furlink incorporates multiple layers of security and integrity controls through a multi-tiered approach covering authentication, authorization, payment security, storage, data integrity, and operational continuity.

- **Authentication and Authorization:** All access is controlled through Supabase Auth using JWT. Upon login, a signed token is issued and scoped to the user's role, expiring after a defined period. Role-Based Access Control (RBAC) restricts each user type to their permitted modules and operations, while Row Level Security (RLS) enforces the same boundaries independently at the database level. Suspended or deactivated users are denied access regardless of token status.

- **Input Validation:** Validation rules are enforced across all forms to prevent malformed or malicious data from entering the system. These rules cover required fields, data types, character limits, and format constraints, and are documented in the project's test cases.
- **Payment Security:** furlink uses PayMongo's hosted checkout page, ensuring raw card details are never collected, stored, or processed by furlink. All financial data is handled within PayMongo's PCI-compliant environment, with booking finalization triggered only upon receipt of a verified webhook signal. The AI Booking Assistant is strictly limited to generating a PayMongo checkout link and cannot access or store any card or wallet information.
- **AI Security:** The Gemini API key is stored exclusively as a Supabase secret and is accessed only within Edge Functions. It is never exposed to the client-side application. The AI Booking Assistant is restricted by the system instructions to only access data the user has explicitly provided or previously stored in their profile, in compliance with the Data Privacy Act of 2012.
- **Storage Security:** Pet media files are stored in Supabase Storage and are planned for migration to private bucket access, where files will only be accessible through time-limited signed URLs tied to authenticated user sessions.
- **Account Integrity:** Account deactivation uses a soft-delete approach, flagging the account as inactive and revoking access while retaining data for audit purposes. Full data removal is available upon a formal erasure request in compliance with the Data Privacy Act of 2012 (RA 10173).
- **Audit Trail:** Login events are recorded in the `user_sessions` table, and database triggers across key tables log all significant data modifications, providing a traceable record of system activity for administrator review.
- **Operational Continuity:** furlink relies on Supabase and Vercel's built-in redundancy, automatic failover, and regular backup capabilities to protect against infrastructure-level failures and maintain platform availability.

Sponsor Acceptance

Approved by the Project Sponsor:



Manuel R. Doria Jr.
Project Sponsor Representative

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